

# ADTA 5750: Applied Natural Language Processing

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## Final Project

### 1. Overview

The final project covers all the topics discussed during the course. The materials in any format posted for the class activities should be considered and used for the project. Additionally, the student can use any other source of information that they can gather.

#### **IMPORTANT NOTES:**

- ) *The student should present their work for each section using both text and images.*
- ) *The sources can be from class lectures, assignments, etc., or other sources*
- ) *One picture is worth 1000 words. Adding one or more images along with the text would be much more convincing. However, an image without text explaining what it is and what it is for is considered incomplete.*
- ) *Images can include the screenshots that the student has taken while working on the classwork.*

#### **IMPORTANT NOTES:**

- ) *When discussing a topic or answering a question, the student is expected to provide adequate explanation and supporting details to his/her presentation.*

#### **IMPORTANT NOTES:**

- ) *If MS Docx is the document format required for submission, the student is required to submit the contents as MS Docx files, **not** submit PDF documents.*
- ) *However, before submitting, the student should **make a backup copy of the documents** by converting them into PDF files that could be used for re-submission if the submitted file is corrupted.*

### 2. Final Project: Assignment Format

The final project is assigned as a group assignment. It means that all the student members of a group will collaborate with each other while working on the assignment.

However, each student is required to **write** and **submit** his/her report **independently**. In other words, a student works on the assignment with the group but **writes** and **submits** the report **as if he/she had worked on the assignment independently**.

### 3. PART I: Final Project: Data Model Draft Presentation (10 Points)

As discussed, HW 4 and HW 5 are part of the final project. In HW 4, the data model of the conversational AI system is most significant, not only for HW 4 but also for the entire final project.

The topic “Data Model” of the AI Dialogue system was discussed at length in the WEEK-10 class meeting. Each group should ensure that they capture the concept and the design of the system’s data model correctly because the quality of the data model significantly impacts the quality of the conversational AI system that the group aims to develop.

#### WHEN:

Data Model Draft Presentation: **Monday, 10/28/2024 (WEEK 11).**

#### WHAT:

The presentation shows only **ONE** PowerPoint slide that includes:

1. The **name** of the AI Dialogue System
2. The **list of entity types**, each of which is shown with some entities

#### HOW:

- Each group has **5 minutes** for the presentation.
  - The PowerPoint slide **must be saved** on a USB drive and plugged into the classroom PC to make the presentation.
- To **receive grades** for the presentation, **every group member** must **participate** in the presentation, i.e., actively deliver part of the presentation.

### 4. PART II: Project Planning, System Analysis, Midway Presentation (10 Points)

#### 4.1 Business, Technical, and Data Requirements

##### SUBMISSION REQUIREMENT PART II #1:

--) Write a report to document the project’s business, technical, and data requirements, i.e., the data model, as the student has completed in Homework Assignment 4.

#### 4.2 Conversational AI System: System Analysis

##### SUBMISSION REQUIREMENTS PART II #2:

--) Write a report to perform the project analysis on planning and feasibility of the project as the student has completed in Homework Assignment 5.

### 4.3 Final Project: Midway Presentation

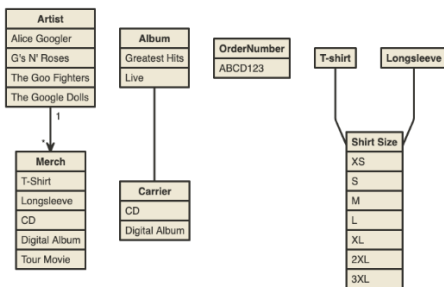
Each group **presents its work on HW 4** in the **Midway Final Project Presentation** on **Monday, 11/04/2024 (WEEK 12)**.

The presentation focuses on two questions:

3. What conversation AI system does the group develop for the final project?
4. What is the data model of the AI Dialogue system?

**NOTES:** *The data model lists all entities and entity types, including both user-defined and system-entity types.*

Based on the documentation of data requirements of the system (done in HW 4), the student is required to design the data model. The work should produce data diagrams like these:



For each entity type, the student must describe it in detail and provide all related information about each entity belonging to the entity type.

#### **IMPORTANT NOTES:**

--> *For each group, every member must participate in the presentation, i.e., actively deliver part of the presentation, so that he/she can get grades for PART I.*

- *Any group member who did not present any content of the group presentation would not get any points for PART II.*

## 5. PART III: Conversational AI System: System Design (30 Points)

Based on the work done in PART I, PART II, PART III, HW 3, and the midterm, the student designs the following major components of the conversational AI system:

- Virtual Agent
- Flows
- Intents
- State Machine
- Pages & State Handlers
- Conditional Responses (if necessary)

#### **SUBMISSION REQUIREMENTS PART III:**

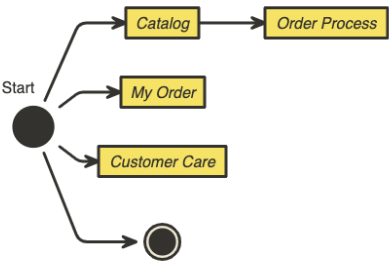
--> Submit an MS Word document, "**ADTA\_5750\_final\_project\_design.docx**," that includes all the above required submitted diagrams and text documentation.

### 5.1 Virtual Agent

The student must describe in detail the purposes for which the virtual agent should be created and all relevant information, including its name.

### 5.2 Flows

The student must determine which flows of dialogues are supported by the system. The work should produce a diagram like this (HW 3 and the midterm):



For each flow, the student must document in detail what the flow is required for the system.

### 5.3 Intents

The student must design all the intents that can be matched to start dialogues using the conversational AI system. The work should produce a table like this:

| Display name              | Training phrases                                                                                                                                                                                                                                                 |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| redirect.product.overview | "Which products do you sell?", "What merchandise items do you have?", "What are you selling?", "What are the items?", "Which products?" "What merchandise?", "Please tell me what you have"                                                                      |
| confirm.artists.overview  | "Yeah, let me buy merchandise", "Yes, I want to purchase something", "Yes, I would like to order merchandise from Alice Googler" (Note: Alice Googler, should be recognized as an @Artist entity!), "Ok, let's buy stuff."                                       |
| redirect.price            | "How much does a t-shirt cost?", "What's the price for the tour movie?", "The album is how much?", "I want to know the price of a longsleeve shirt", "What's the price difference?", "What does each product costs?", "What does it cost?", "What is the price?" |

For each intent, the student should provide the following pieces of information:

- (1): Intent name
- (2): General description of what the speaker tries to do with the intent
- (3) The training phrases of the intent (at least 10 training phrases)

### 5.4 State Machine

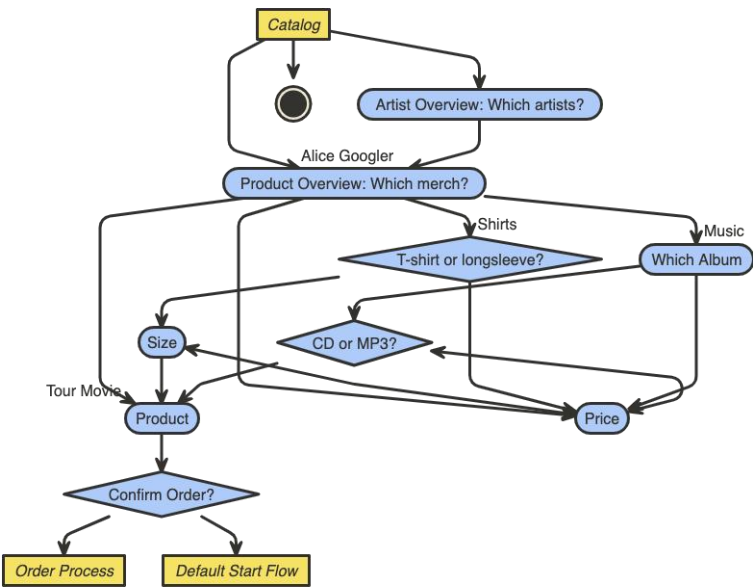
...

5.5 Pages and State Handlers

Based on the intents, the student must design the pages of the system. The work should produce a table of page intent routes like this:

| Page (In Flow) | Routes > Intent            | Routes > Transition To |
|----------------|----------------------------|------------------------|
| Start          | Default Welcome Intent     | -                      |
| Start          | redirect.artists.overview  | Flow: Catalog          |
| Start          | redirect.product.overview  | Flow: Catalog          |
| Start          | redirect.shirts            | Flow: Catalog          |
| Start          | redirect.music             | Flow: Catalog          |
| Start          | redirect.product           | Flow: Catalog          |
| Start          | redirect.product.of.artist | Flow: Catalog          |
| Start          | redirect.refund.info       | Flow: Customer Care    |
| Start          | redirect.shipping.info     | Flow: Customer Care    |
| Start          | redirect.swapping.info     | Flow: Customer Care    |
| Start          | redirect.my.order          | Flow: My Order         |
| Start          | redirect.my.order.cancel   | Flow: My Order         |
| Start          | redirect.my.order.status   | Flow: My Order         |
| Start          | redirect.end               | Page: End Session      |

The student is also required to submit the state machine of the system which includes pages like this:



5.6 Conditional Responses (If Necessary)

The student should design necessary conditional responses, following the steps shown in the documentation of the codelab project (GG-Records).

## 6. PART IV: Conversational AI System: Implementation and Development (20 Points)

Based on the above system design, the student must develop the conversational AI system using GCP Dialogflow CX and document in detail every step along the process.

For each step, the student should document the following details:

- (1): Describe in detail what they have done
- (2): Capture related screenshots to illustrate what they have done, including work results after completing the step.

### SUBMISSION REQUIREMENTS PART IV:

--> Submit an MS Word document, “[ADTA\\_5705\\_final\\_project\\_system\\_development\\_report.docx](#),” that includes the details and screenshots of all steps of the process.

## 7. PART V: Conversational AI System: Testing and Deployment (10 Points)

The student must test the conversational AI system developed above and document every step of the process.

In the document, the student must report in detail what they have done to test the conversational AI system, including:

- The tools or subsystems of GCP Dialogflow CX that are used to test the system
- Each complete dialogue between the virtual agent, and the imagined customer illustrates the features of the system
- Screenshots taken at each step

While testing the system, the student must test all the flows and possible scenarios the state machine shows.

### SUBMISSION REQUIREMENTS PART V:

--> Submit an MS Word document, “[ADTA\\_5705\\_final\\_project\\_system\\_testing\\_report.docx](#),” that includes the details and screenshots of all process steps.

## 8. PART VI: Final Project: Presentation (20 Points)

Each group **will present the AI dialogue system** that the group has developed for the final project on **Monday, 12/02/2024 (WEEK 16)**.

To get grades for PART VI, each member of each group **must participate in the presentation**, i.e., actively deliver part of the presentation, to get grades for PART V.

- Any group member who did not present any content in the group presentation would not get any points for PART V.

**IMPORTANT NOTES:**

*--> All groups must be in class by the presentation's start.*

*--> To be fair, the order of presenting, i.e., which group makes the presentation first, second, and so on, is determined by drawing a group number randomly.*

## **9. HOWTO Submit**

### **9.1 Final Project Report and All Related Documents**

The student must submit all required documents (MS Word documents) for the final project by uploading them to their OneDrive folder for submissions.

**IMPORTANT NOTES:**

*--> The student receives a link to access the folder via the UNT email.*

The following documents are required for the final project reports:

1. Project: Business, Technical, and Data Requirements (PART II: **HW 4**)
2. Conversational AI System: System Analysis (PART II: **HW 5**)
3. Conversational AI System: System Design (PART III)
4. Conversational AI System: Implementation & Development (PART IV)
5. Conversational AI System: Testing and Deployment (PART V)

The student must inform the instructor about the submission by email sent to the instructor ([Thuan.Nguyen@unt.edu](mailto:Thuan.Nguyen@unt.edu) )

The subject of the email must be: “**ADTA 5750: Final Project – Submission.**”

**Due date & time: 8:00 AM – Monday 12/02/2024**

**IMPORTANT NOTES:**

*--> Due to the limited time required to grade and post the grades as required by the Registrar's Office, **no late submission** is accepted.*